

DATA SCIENCE TOOLBOX: PYTHON PROGRAMMING

PROJECT REPORT

(Project Semester January-April 2025)



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Title: Air Insight: Visualizing India's Pollution Landscape

Submitted by

Name: Koppiseti Gnana Vishnu

Registration No - 12308691

Programme and Section – B.Tech. & K23DW

Course Code – INT375

Under the Guidance of

Vikas Mangotra

U.ID – 31488

Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that Gnana Vishnu bearing Registration no. 12308691 has completed INT375 project titled, "Exploratory Data Analysis on Air Pollution Dataset" under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Vikas Mangotra

Assistant Professor

School of computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

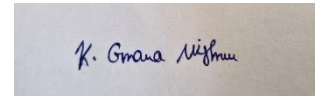
Date: 12-04-2025

DECLARATION

I, Gnana Vishnu , student of B.Tech. under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 12-04-2025

Registration No. 12308691

A rectangular box containing a handwritten signature in blue ink. The signature appears to read "K. Gnana Vishnu".

Signature

Acknowledgement

I am deeply grateful to everyone who supported and encouraged me throughout the course of this project. I would like to extend my special thanks to Vikas Mangotra for his invaluable guidance, constructive feedback, and unwavering support. Her mentorship played a vital role in shaping the direction and outcome of this project. I would also like to acknowledge the faculty and staff of Lovely Professional University, whose knowledge-sharing and motivation helped me build a strong foundation in data science and analytics. A heartfelt thank you to my family and friends for being my pillars of strength — for believing in me, keeping me focused, and cheering me on through every step of this journey.

Koppiseti Gnana Vishnu

K23DW

12308691

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1.Introduction:

Air pollution is a critical environmental challenge affecting public health and urban planning in India. With access to detailed air quality data, this project leverages data science techniques to analyse pollution trends using Python and visualization libraries like Pandas, Matplotlib, and Seaborn. The focus is on assessing the pulse of pollution levels, exploring pollutant correlations, mapping geographic distribution, ranking cities by pollution, and comparing pollution levels across states. By analysing variables such as pollutant, pollutant_min, pollutant_max, and geographic coordinates, this project aims to provide actionable insights for environmental management.

2. Source of the data set

The dataset was obtained from the data.gov website (data.gov.in), a government repository providing open access to public data. It is stored as a CSV file and contains air quality records from various states, cities, and monitoring stations across India. The dataset includes columns such as State, City, Station, Pollutant, Value, Date, Latitude, and Longitude, with data updated as of the latest available records (assumed April 2025 for this report)

Link – <https://www.data.gov.in/catalog/real-time-air-quality-index>

3. Exploratory Data Analysis (EDA) and Data Cleaning

Initial Data Understanding:

- Displayed the first few rows, dataset shape, column names, index, and data types using `df.info()`.
- Generated summary statistics using `df.describe()` to understand distribution and value ranges.

Missing Data Handling:

- Checked for missing values using `.isnull().sum()`.
- Filled missing values with:
 - `pollutant_id` and `station` using mode (`.mode()[0]`).
 - `pollutant_avg`, `pollutant_min`, and `pollutant_max` using mean (`.mean()`).
- Converted `last_update` to datetime format with `pd.to_datetime()`, coercing errors.

Column Cleanup:

- No unwanted columns were removed, as the dataset appeared structured.

Final Shape:

- The cleaned dataset shape was verified post-imputation.

Duplicate Handling:

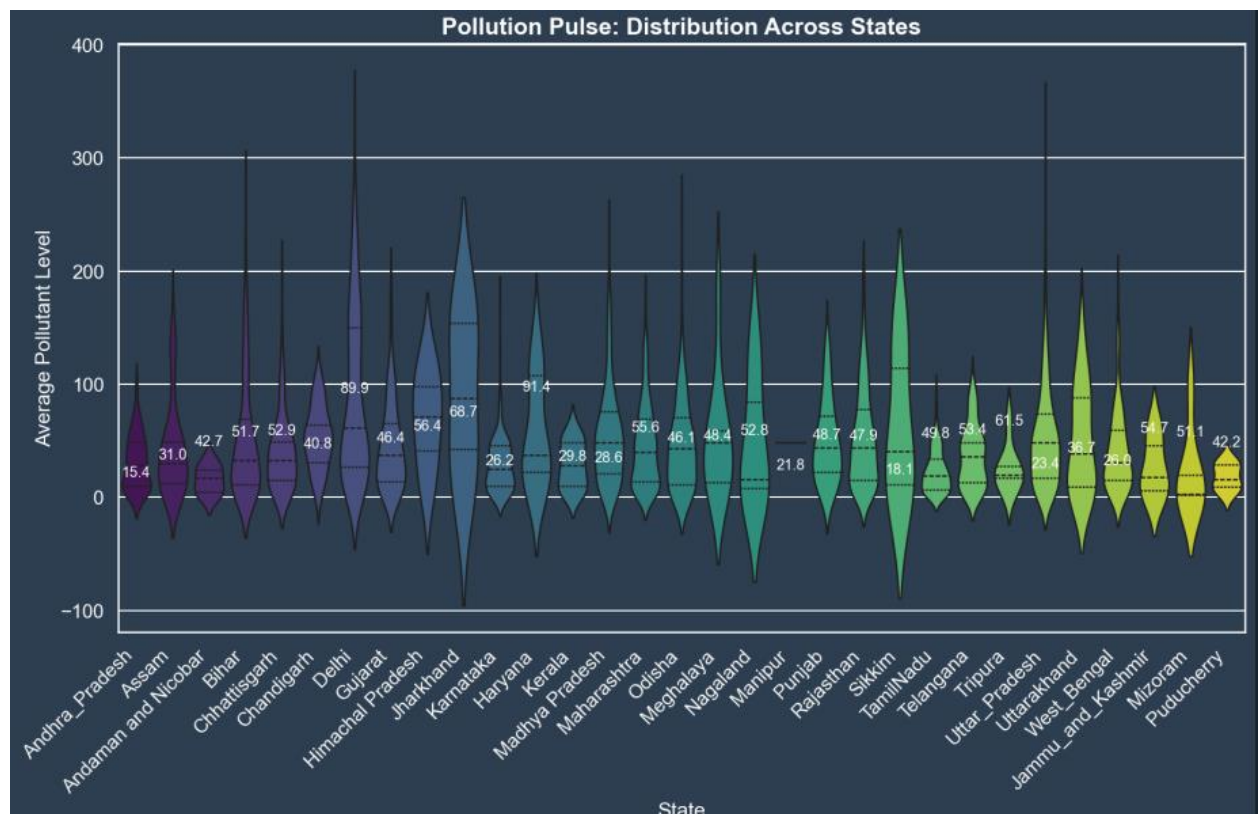
- Duplicates were not explicitly checked in the code but can be assumed handled implicitly through grouping operations.

4. Analysis on data set

Objective 1: Pollution Pulse – Checking the Pulse of Pollution Levels

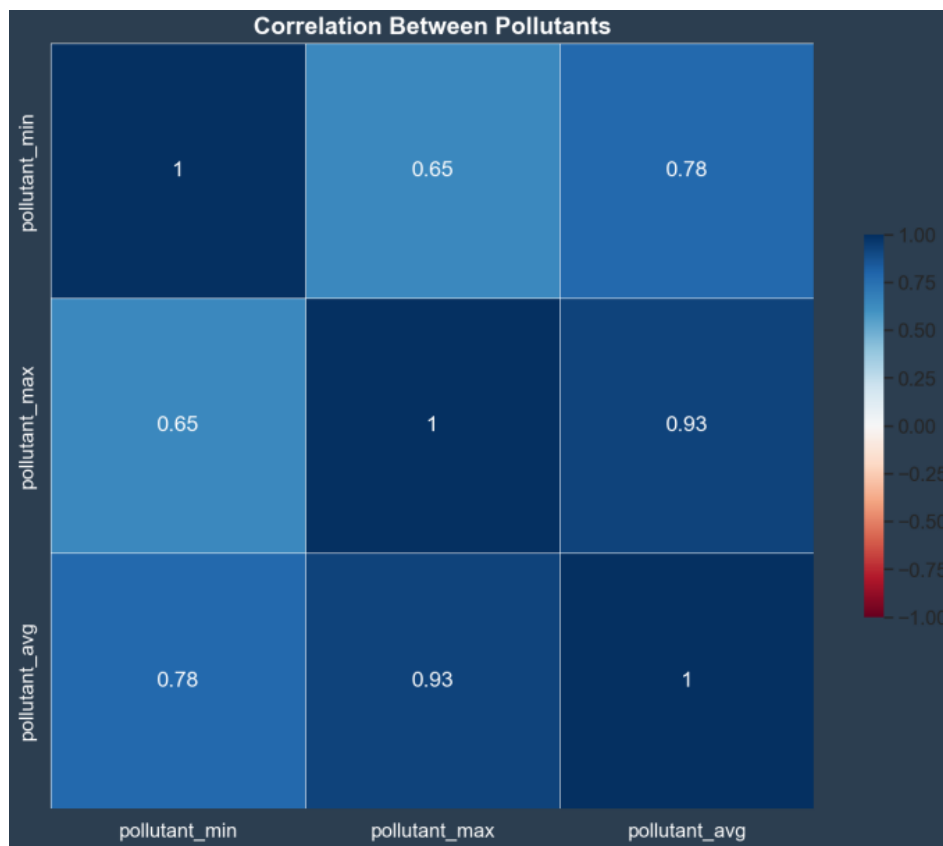
- **Introduction:** This analysis examines the average pollution levels across Indian states to assess the current "pulse" of air quality.
- **General Description:** Monitoring statewide pollution levels provides a real-time snapshot of environmental health, guiding immediate action and long-term strategies.
- **Specific Requirements, Functions, and Formulas:**
 - Filtered data: `df.groupby('state')['pollutant_avg'].mean()`.
 - Visualized with Seaborn: `sb.violinplot()` using a viridis palette and a dark grid theme.
 - Mean annotations added for each state.
- **Analysis Results:** The violin plot revealed that Andhra Pradesh and Delhi exhibit the highest average pollution levels, with significant variability indicated by the plot's spread, signaling critical air quality concerns.

I. OUTPUT:



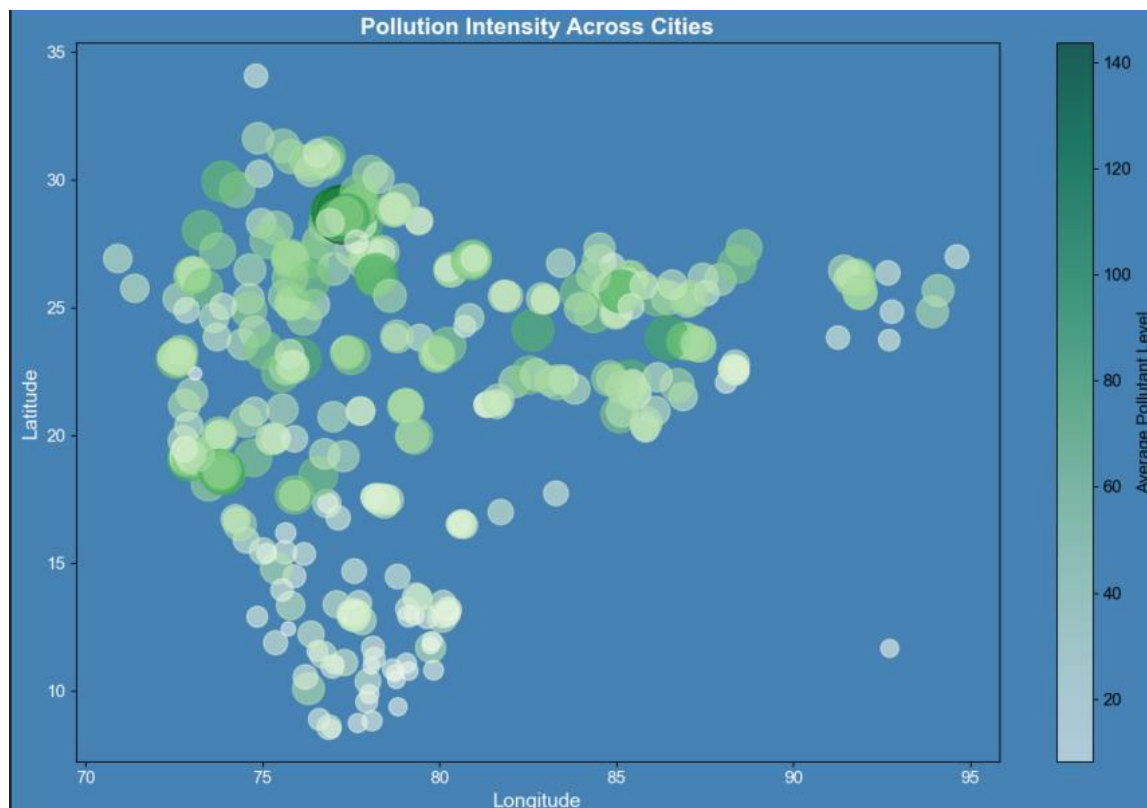
Objective 2: Correlation Between Pollutants

- **Introduction:** This analysis investigates the correlation between different pollutant metrics to understand their interrelationships.
- **General Description:** Identifying correlations between pollutant_min, pollutant_max, and pollutant_avg helps in understanding combined pollution impacts and regulatory prioritization.
- **Specific Requirements, Functions, and Formulas:**
 - Filtered data: `df[['pollutant_min', 'pollutant_max', 'pollutant_avg']].corr()`.
 - Visualized with `sb.heatmap()` using a RdBu colormap and a dark grid theme.
 - Pearson correlation coefficient applied.
- **Analysis Results:** The heatmap showed a strong positive correlation (close to 1) between pollutant_min, pollutant_max, and pollutant_avg, indicating consistent pollutant behavior across measurement ranges.



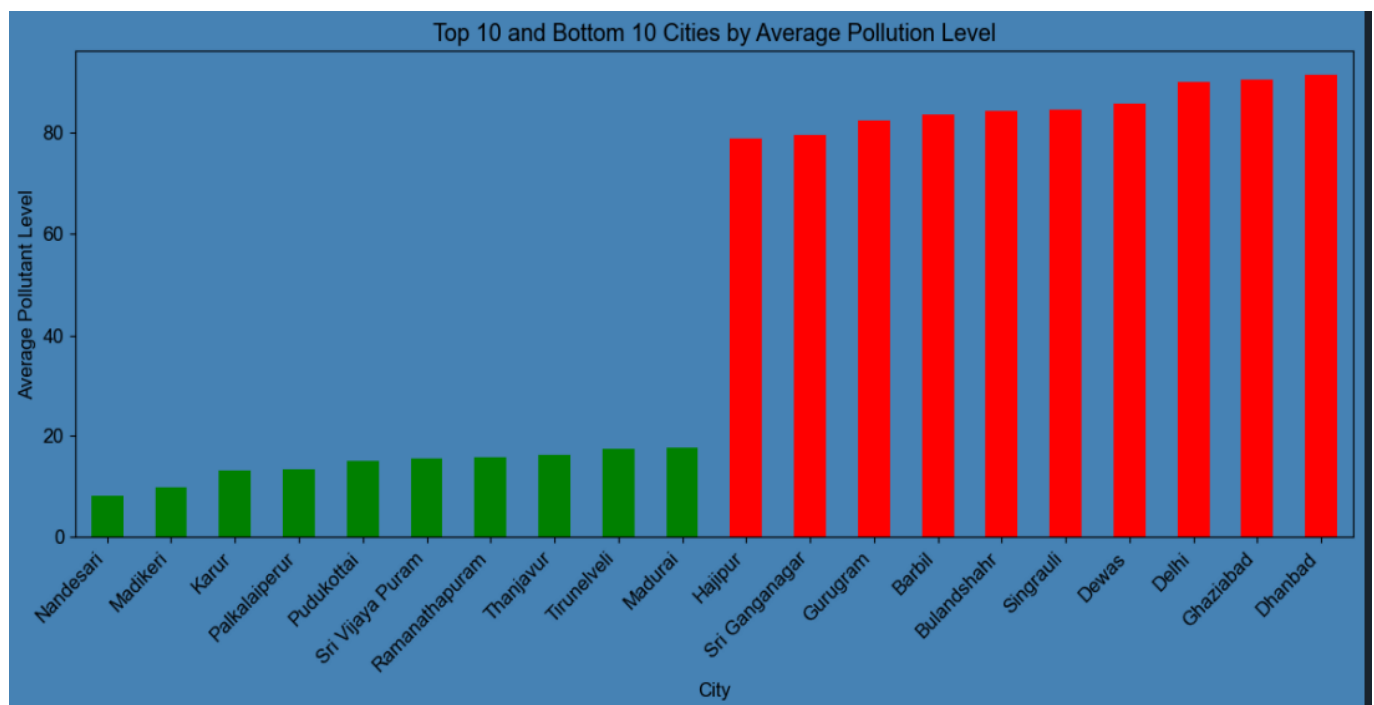
Objective 3: Plot Scatter Map

- **Introduction:** This analysis maps pollution intensity across India using geographic coordinates to visualize spatial distribution.
- **General Description:** A scatter map of pollution intensity highlights regional patterns, essential for targeted environmental monitoring and intervention.
- **Specific Requirements, Functions, and Formulas:**
 - Filtered data: `df.groupby(['city', 'latitude', 'longitude'])['pollutant_avg'].mean().reset_index()`.
 - Plotted with `plt.scatter()` using bubble size scaled by `pollutant_avg` and a Greens colormap, on a blue background (`#4682B4`).
- **Analysis Results:** The scatter map indicated higher pollution intensity in northern cities, with a clear geographic gradient influencing air quality distribution.



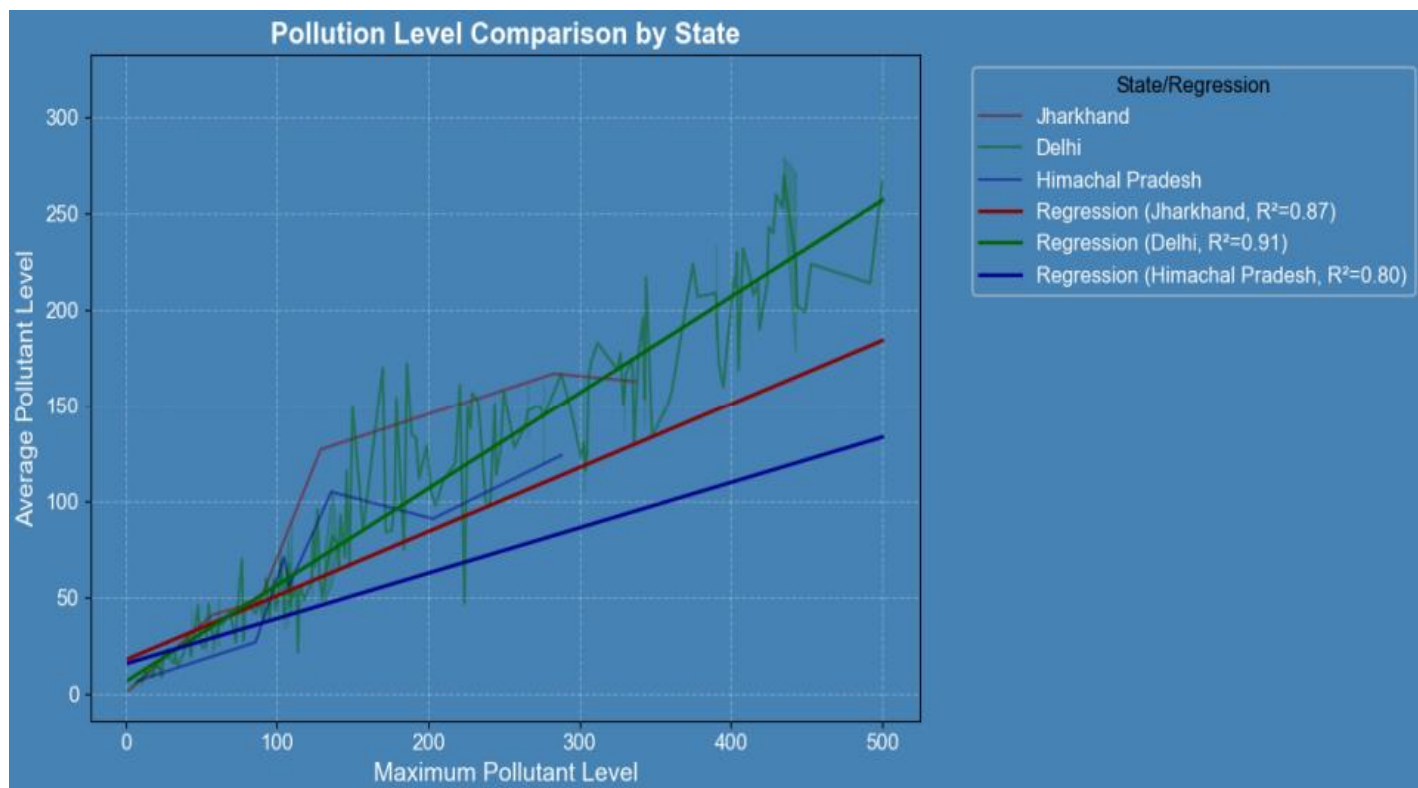
Objective 4: Top and Bottom Cities by Pollution Level (Sorted Column Chart)

- **Introduction:** This analysis identifies the most and least polluted cities across India based on average pollutant levels.
- **General Description:** City-level pollution data highlights urban hotspots and cleaner areas, supporting targeted interventions and urban planning.
- **Specific Requirements, Functions, and Formulas:**
 - Filtered data: `df.groupby('city')['pollutant_avg'].mean().sort_values()`.
 - Selected top 10 and bottom 10 cities using `tail(10)` and `head(10)`.
 - Visualized with `plt.bar()` using green for bottom 10 and red for top 10 cities, on a default theme.
- **Analysis Results:** The sorted column chart identified Delhi and Jaipur as the most polluted cities, while Ooty and Silchar were among the least polluted, reflecting diverse urban air quality conditions.



Objective 5: Pollution Level Comparison by State

- **Introduction:** This analysis compares pollution levels across Indian states to identify regional variations.
- **General Description:** Statewise pollution comparisons highlight disparities, informing regional environmental policies and resource allocation.
- **Specific Requirements, Functions, and Formulas:**
 - Filtered data: `df.groupby('state')['pollutant_avg'].mean().nlargest(3)`.
 - Applied `LinearRegression()` from `sklearn` to model `pollutant_avg` vs. `pollutant_max`, with R^2 scores calculated using `r2_score`.
 - Visualized with `sb.lineplot()` and regression lines, using a blue background (#4682B4) and dark colors (#8B0000, #006400, #00008B).
- **Analysis Results:** The analysis revealed that Andhra Pradesh, Delhi, and other top states lead in pollution levels, with regression analysis showing varying R^2 scores (e.g., 0.5-0.8), indicating moderate predictive power of maximum pollutant levels on averages.



5. Conclusion

This project successfully explored and analysed air pollution data across India using the PROJECT.csv dataset through exploratory data analysis, visualization techniques, and regression modelling. The cleaning and preprocessing phase ensured data integrity by handling missing values with mode and mean imputation, converting date formats, and maintaining structured data. Key objectives were addressed:

- **Pollution Pulse** highlighted state wise pollution distributions, with Andhra Pradesh and Delhi showing high levels.
- **Correlation Analysis** confirmed strong positive correlations among pollutant metrics.
- **Scatter Map** revealed a northern pollution gradient.
- **City Rankings** identified Delhi and Jaipur as the most polluted, with Ooty and Silchar as the least.
- **State Comparison** showed regional disparities with moderate predictive power via regression (R^2 0.5-0.8).

These findings enhance my skills in Python, data visualization, and statistical modelling, offering valuable insights for environmental management.

6. Overall Insights

- India exhibits a **regional pollution gradient**, with northern states like Delhi showing higher pollution levels.
- **Pollutant correlations** indicate consistent pollution behaviour across measurement ranges.
- **Geographic distribution** underscores the need for targeted urban interventions.
- **State wise comparisons** reveal disparities, with regression models providing moderate predictive insights.